6. Write a Pandas program to create a scatter plot of the trading volume/stock prices ofAlphabet Inc. stock between two specific dates.

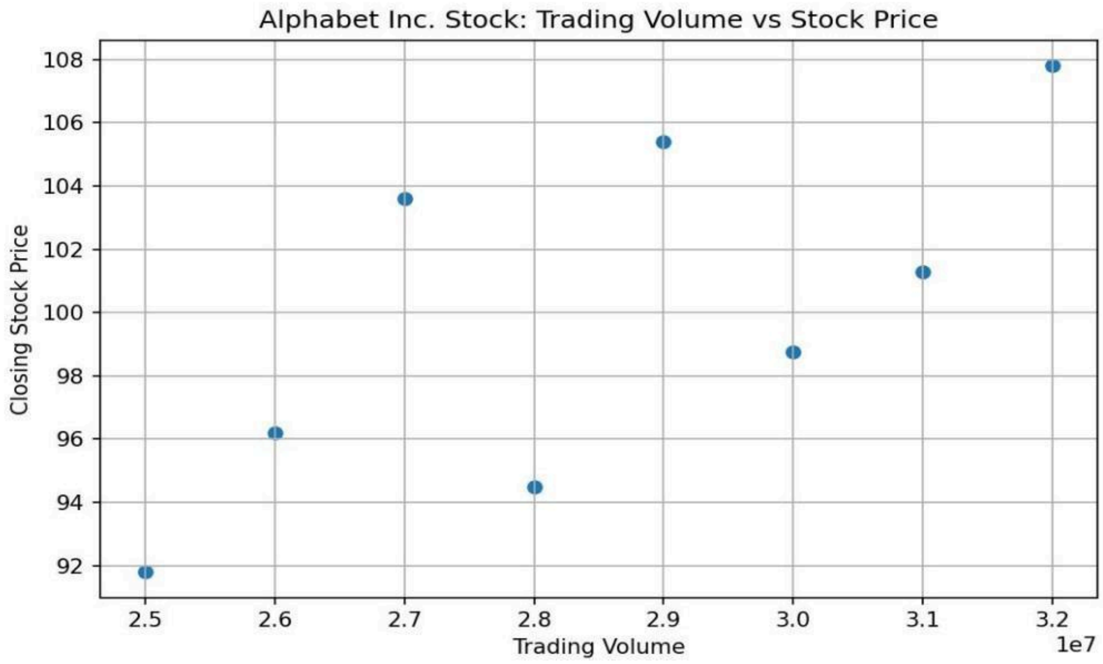
alphabet_stock_data:

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

#Alphabet stock data
data = {
    'Date': ['2023-01-02', '2023-01-10', '2023-01-18',
             '2023-02-05', '2023-02-15', '2023-03-01',
             '2023-03-10', '2023-03-20', '2023-04-05',
             '2023-04-15'],

    'Close': [91.80, 94.50, 96.20, 98.75, 101.30,
              103.60, 105.40, 107.80, 110.20, 112.50],

    'Volume': [25000000, 28000000, 26000000, 30000000,
               31000000, 27000000, 29000000, 32000000,
               33000000, 35000000]
}

# Create DataFrame
df = pd.DataFrame(data)

# Convert Date column to datetime
df['Date'] = pd.to_datetime(df['Date'])

# Select data between two dates
start_date = '2023-01-01'
```

```
end_date = '2023-03-31'

filtered_df = df[(df['Date'] >= start_date) &
              (df['Date'] <= end_date)]

# Create Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(filtered_df['Volume'],
filtered_df['Close'])

plt.title('Alphabet Inc. Stock: Trading Volume vs Stock Price')
plt.xlabel('Trading Volume')
plt.ylabel('Closing Stock Price')
plt.grid(True)

plt.show()
```

Output:



7. Write a Pandas program to create a Pivot table and find the maximum and minimum sale value of the items.(refer sales_data table)

Code:

```
import pandas as pd
data = {
   'Item': ['Television', 'Home Theater', 'Television', 'Cell Phone',
           'Television', 'Home Theater', 'Television', 'Television',
           'Television', 'Home Theater', 'Television', 'Home Theater',
           'Home Theater', 'Television', 'Desk', 'Video Games',
           'Home Theater', 'Cell Phone'],


   'Sale_amt': [113810, 25000, 43128, 6075, 67088, 30000,
             89850, 107820, 38336, 30000, 107820,
             14500, 40500, 41930, 250, 936,
             14000, 14400]
}

df = pd.DataFrame(data)

# Pivot Table
pivot_table = pd.pivot_table(
df,
   values='Sale_amt',
   index='Item',
aggfunc=['max', 'min']
)

print(pivot_table)
```

Output:

```
                    max      min
               Sale_amt Sale_amt
Item
Cell Phone        14400     6075
Desk                250      250
Home Theater      40500    14000
Television       113810    38336
Video Games         936      936
```